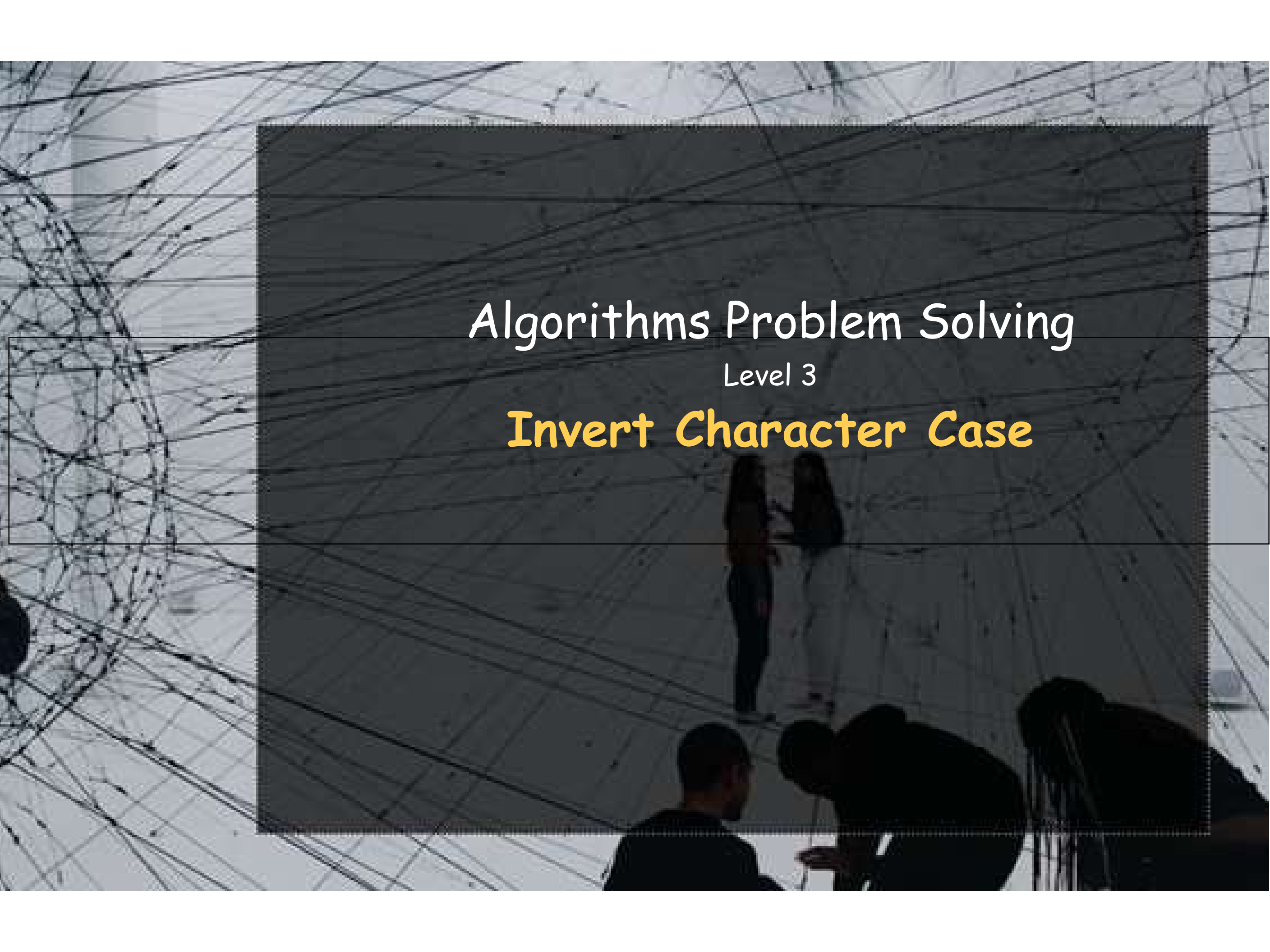


Algorithms Problem Solving

Level 3

Invert Character Case

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the foreground, the silhouettes of several people are visible, some standing and some sitting, suggesting a collaborative workspace or a public area. The overall tone is professional and tech-oriented.

Algorithms Problem Solving

Level 3

Invert Character Case

Definition

Single Character Case Toggle

 Program switches one character from uppercase to lowercase or from lowercase to uppercase.

 Non-alphabetic characters remain unchanged.

Explanation

Case Detection

-  ``isupper()`` is used to determine whether the character is uppercase.

-  The program can also explicitly check ``islower()`` when a stricter implementation is desired.

Conditional Conversion

-  Uppercase characters are converted using ``tolower()``.

-  Lowercase characters are converted using ``toupper()``.

Character Return

-  The converted character is returned to the caller.

-  The function can then be reused by larger string-processing functions.

Procedures

- 1 Read a character.
- 2 Determine its current case.
- 3 Convert it to the opposite case.
- 4 Return and display the resulting character.

💡 Example

Input: a

→ Result
A

Input: A

→ Result
a

Note

- ◆ Digits and punctuation are unaffected by ``toupper()`` and ``tolower()``.
- ◆ The function is a reusable helper for full-string case inversion.
- ◆ ``cin >> ch`` skips whitespace characters.

⊘ Warning

- ◆ Checking only ``isupper()`` means non-uppercase characters enter the alternative branch; this is safe for typical ASCII letters but explicit ``islower()`` checking can make the intention clearer.
- ◆ Character classification functions from ``<cctype>`` require appropriate character handling.
- ◆ The return type should match the expected character type.



Comparison

▨ `InvertLetterCase()`

- ✓ Changes uppercase to lowercase and lowercase to uppercase.
- ✓ Requires conditional logic based on the current case.
- ✓ Performs a bidirectional transformation.

▤ `toupper()` / `tolower()`

- ✓ Forces a character toward one specific case.
- ✓ Does not automatically switch based on the original case.
- ✓ Represents one-directional case conversion.

✦ Summary

- ✓ Inspects the case of one character.
- ✓ Converts uppercase letters to lowercase.
- ✓ Converts lowercase letters to uppercase.
- ✓ Leaves non-alphabetic characters unchanged.

Thank you 😊